THERMODYNAMICS

Heat Motion

→ Heat transfer.

→ Work transfer.

Heat ↔ Work.

Classical thermodynamics → Macroscopic Study.
Statistical thermodynamics → Molecular level study.

State 1 → state 2. condition

Thermodynamic Equillibrium.

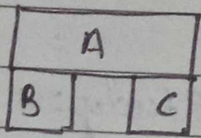
Thermal Eqm   Mechanical Eqm   chemical Eqm.

control mass approach —.

→ Temperature is associated with the ability to distinguish hot from cold.

ZEROTH LAW OF THERMODYNAMICS – It is basis of temp. measurement

When a body A is in thermal Equillibrium with a body B, and so separately with a body C. then bodies B & C will be in thermal Eqm with each other.



temperature measurement (Reference body)

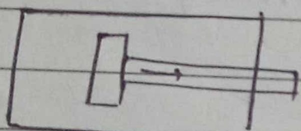
$T_A = T_B$

$T_B = T_C$

$\boxed{T_A = T_C}$

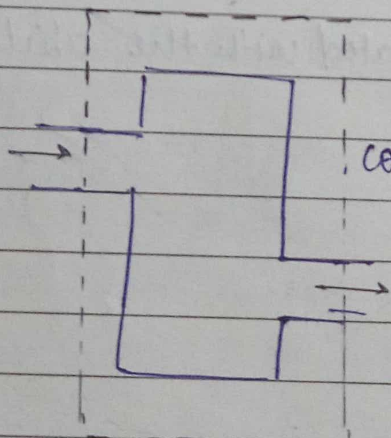
Thermometer → Mercury → thermomatric property → change with temp.

① $\int Pdv$ → displacement work. (Reversible/slowly)



$P \times dl \times A \to \int Pdv$

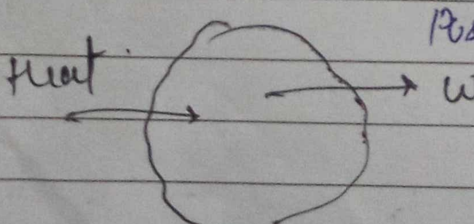
② flow work.



$P_1 V_1$ or

$m, P_1 v_1$ → specific volume.

Sign convention –



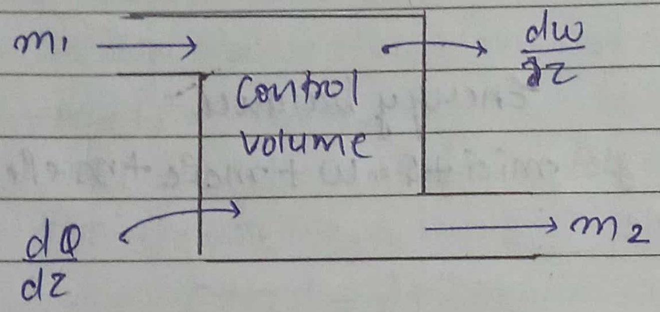
Heat supply to system → positive

work done by system → positive

based on

FIRST LAW OF THERMODYNAMICS – Conservation of Energy.

→ for cycle process → $\oint Q = \oint W$
→ change of state → $\boxed{Q = E_2 - E_1 + W}$ } stationary system.



} steady flow process
↳ no change in property wrt time.

$$\dot m_1 e_1 + \frac{dQ}{dz} = \frac{dW}{dz} + \dot m_2 e_2 + m_2 P_2 v_2 - m_1 P_1 v_1$$

↳ $\left(u + \frac{V^2}{2} + gz\right)$

→ mass balance. $\dot m_1 = \dot m_2$

→ Energy balance

$$m_1\left(u_1 + \frac{V_1^2}{2} + gz_1\right) + \frac{dQ}{dz} = \frac{dW}{dz} + m_2\left(u_2 + \frac{V_2^2}{2} + gz_2\right) + m_2 P_2 v_2 - m_1 P_1 v_1$$

$\boxed{h = u + pv}$

↳ $$m_1\left(h_1 + \frac{V_1^2}{2} + gz_1\right) + \frac{dQ}{dz} = \frac{dW}{dz} + \dot m_2\left(h_2 + \frac{V_2^2}{2} + gz_2\right)$$

Unit.

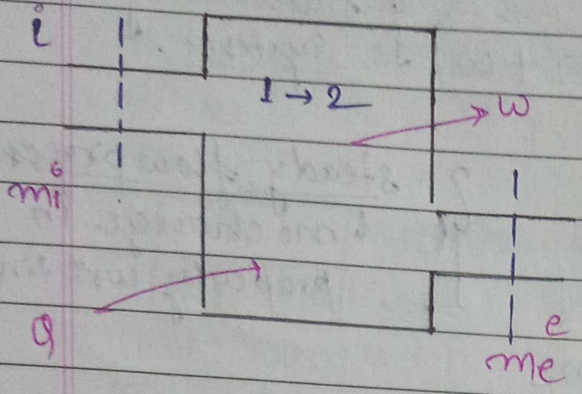
- Heat / work / energy – Joule / Nm.
- $\dot m_1 h_1 = \frac{Kg}{sec} \times \frac{J}{kg} = J/sec.$
- $\frac{m_1 V_1^2}{2} = \frac{Kg}{sec} \times \frac{m^2}{sec^2} = \frac{Kg\text{-}m}{sec^2} \times \frac{m}{sec} = J./sec.$

⇒ we can also write Equation –

$$\left(h_1 + \frac{V_1^2}{2} + gz_1\right) + \frac{dQ}{dm} = \frac{dW}{dm} + \left(h_2 + \frac{V_2^2}{2} + gz_2\right)$$

J/Kg.

unsteady flow Energy Equation -



Mass balance -

$m_i - m_e = m_2 - m_1$

Energy balance.

$m_i e_i + Q = W + m_e e_e + m_e p_e v_e - m_i p_i v_i$

$\Rightarrow m_i e_i + Q = W + m_e e_e + m_e p_e v_e - m_i p_i v_i$

$\Rightarrow m_i\left(h_i + \frac{V_i^2}{2} + g z_i\right) + Q = W + m_e\left(h_e + \frac{V_e^2}{2} + g z_e\right) + m_2\left(u_2 + \frac{V_2^2}{2} + g z_2\right) - m_1\left(u_1 + \frac{V_1^2}{2} + g z_1\right)$

Unit → Joule .

If kinetic and potential Energy neglected

then → $m_i h_i + Q = m_e h_e + W + m_2 u_2 - m_1 u_1$

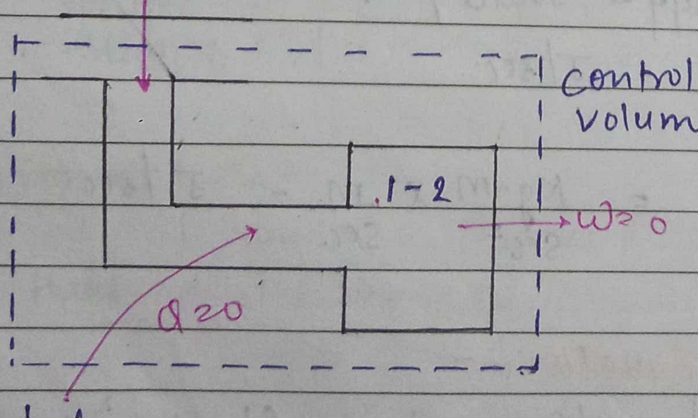
$W = 0$ | $Q = 0$

$m_i h_i = m_e h_e + m_2 u_2 - m_1 u_1$

System initially evacuated ($m_1 = 0$)

$m_i h_i = m_e h_e + m_2 u_2$

Example - filling a bottle from a large source of fluid.



→ mass balance -

$m_i - \cancel{m_e}^{0} = m_2 - m_1$

$m_i = m_2 - m_1$

→ Energy balance

$m_i h_i + \cancel{Q}^{0} = \cancel{m_e h_e}^{0} + \cancel{W}^{0} + m_2 u_2 - m_1 u_1$

$m_i h_i = m_2 u_2 - m_1 u_1$

$(m_2 - m_1) h_i = m_2 u_2 - m_1 u_1$

Initially Evacuated $\boxed{m_1 = 0} \rightarrow \boxed{m_2 = m_i}$

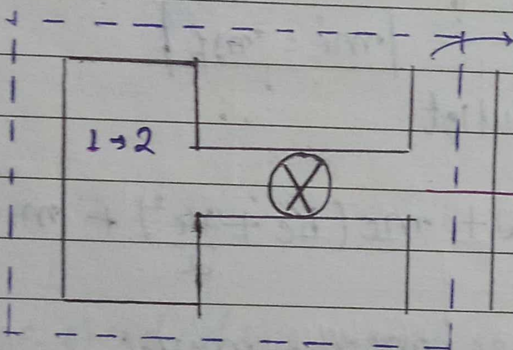
$\Rightarrow m_2 h_i = m_2 u_2$

$\Rightarrow h_i = u_2$ fluid ideal gas → $C_p T_i$

$C_p T_i = C_v T_2$ $\quad C_p T_i = \cancel{C}\ H = m C_p T$

$\boxed{T_2 = \frac{C_p}{C_v} T_i = \gamma T_i}$ $\quad \rightarrow u = m C_v T$

Example → Emptying a Reservior into atmosphere.



mass balance.

$m_i - m_e = m_2 - m_1$

$\boxed{m_e = m_1 - m_2}$

Energy balance

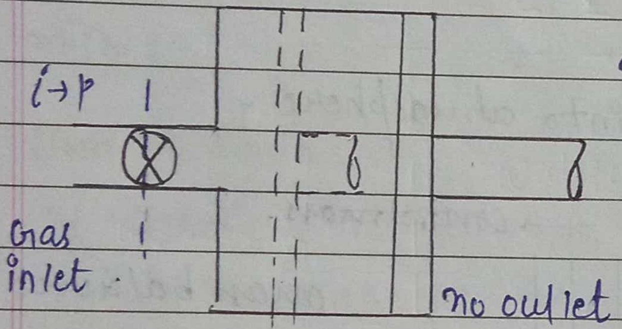
$m_i h_i + Q = m_e h_e + W + m_2 u_2 - m_1 u_1$

$Q = m_e h_e + m_2 u_2 - m_1 u_1 - m_i h_i$

Q In a reciprocating engine the mass of gas occuping the clearance volume is m_c kg at state P_1, u_1, v_1, h_1 by opening the inlet valve m_f kg of gas is taken into the cylinder and at the conclusion of the intake ~~into the cylinder and at the conclusion~~ of process the state of gas is given by P_2, u_2, v_2, h_2 the state of the gas in the supply pipe is constant and is given by P_p, u_p, v_p, h_2 the state of the gas in the supply pipe

Q In a reciprocating engine the mass of gas occupying the clearance volume is m_c kg at state P_1, u_1, v_1, h_1 by ~~occupying~~ opening the inlet valve m_f kg of gas is taken into the cylinder and at the conclusion of the intake process the state of gas is given by P_2, u_2, v_2, h_2 the state of the gas in the supply pipe is constant and is given by P_p, u_p, v_p, h_p & V_p (velocity) How much heat is transfered b/w the gas and the cylinder wall during the intake process.

Solution



mass balance -

$m_i = m_e = m_2 - m_1$

$m_i = m_2 - m_1$

$m_i = m_f + m_c - m_c$

$\boxed{m_i = m_f}$

$m_1 = m_c$ kg

$m_2 = (m_f + m_c)$ kg

$$\Rightarrow Q + m_i\left(h_i + \frac{v_i^2}{2}\right) = W + m_e\left(h_e + \frac{V_e^2}{2}\right) + m_2 u_2 - m_1 u_1$$

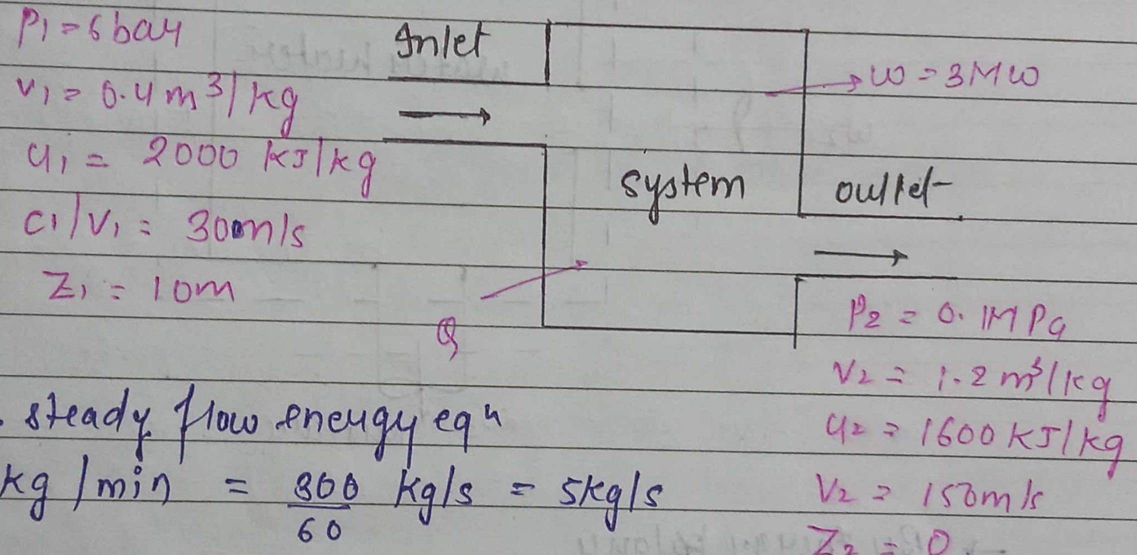
$$\Rightarrow Q + m_p h_p + \frac{m V_p^2}{2} = W + m_2 u_2 - m_1 u_1$$

$$\Rightarrow Q + m_f\left(h_p + \frac{V_p^2}{2}\right) = W + (m_f + m_c) u_2 - m_c u_1$$

$$\Rightarrow Q = W + (m_f + m_c) u_2 - m_c u_1 - m_f\left(h_p + \frac{V_p^2}{2}\right).$$

Q In a steady flow system, a substance flows at the rate of 300 kg/min. It enters at a pressure of 6 bar, a velocity of 300 m/s specific internal energy 2000 KJ/kg and specific volume 0.4 m^3/kg. It leaves the system at a pressure of 0.1 MPa, a velocity of 150 m/s specific, internal energy 1600 KJ/kg and specific volume 1.2 m^3/kg. The inlet is 10 m above the outlet. During its passage through the system the substance has a work transfer of 3MW to the surroundings. Determine the heat transfer in KJ/s. ~~startt~~ stating wheather it is from or to the system.

Solution



Applying the steady flow energy eqn

$\dot{m} = 300$ kg/min $= \frac{300}{60}$ kg/s $= 5$ kg/s

$W = \dot{m}w$

$3000 = 5 \times w$

$w = 600$ KJ/kg.

$$p_1v_1 + u_1 + \frac{V_1^2}{2} + gz_1 + q = p_2v_2 + u_2 + \frac{C_2^2}{2} + gz_2 + w$$

$$(6)(0.4)(100) + 2000 + \frac{(300)^2}{2\times1000} + \frac{9.81\times10}{1000} + q = 100 \times 1.2 + 1600 + \frac{(150)^2}{2\times1000} + 600$$

$q = 46.1519$ KJ/kg

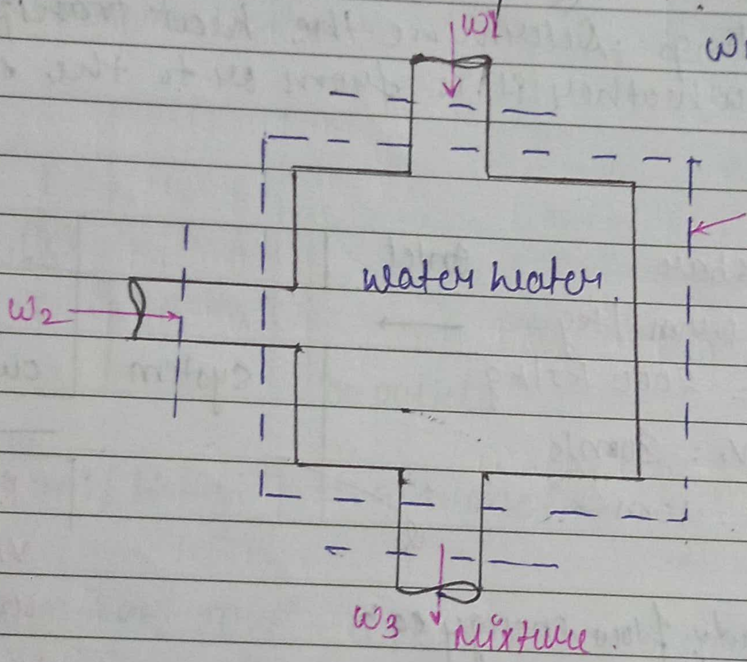
$Q = q \times m$

$Q = 46.1519 \times 5$ KJ/kg $= 230.7595$ KJ/s

Q A certain water heater operates under steady flow conditions receiving 4.2 kg/s of water at 75°C temperature enthalpy 313.93 kJ/kg. The water is heated by mixing with steam which is supplied to the heater at temp 100.2°C and enthalpy 2676 kJ/kg. The mixture leaves the heater as liquid water at temperature 100°C and enthalpy 419 kJ/kg. How much steam must be supplied to the heater per hour?

→ mass balance

$w_1 + w_2 = w_3$



→ By Energy balance.

$$w_1\left(h_1 + \frac{V_1^2}{2} + Z_1 g\right) + \frac{dQ}{dt} + w_2\left(h_2 + \frac{V_2^2}{2} + Z_2 g\right)$$

$$= w_3\left(h_3 + \frac{V_3^2}{2} + Z_3 g\right) + \frac{dW}{dt}$$

potential and kinetic Energy assumed to be Zero

heater → insulated $Q = 0$ & $W = 0$

The Equation Reduce to.

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

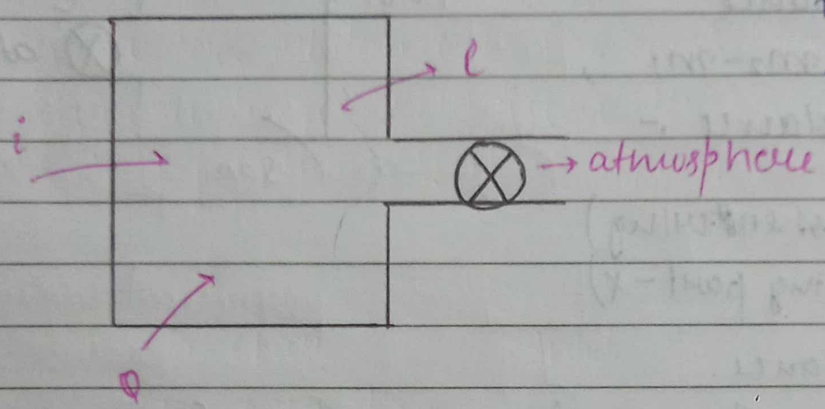
$$4.2 \times 313.93 + w_2 \times 2676 = (4.2 + w_2) 419$$

$$w_2 = 0.196 \text{ kg/s}$$

$$\boxed{w_2 = 705 \text{ kg/hr.}}$$

Q An air receiver of volume $5.5 m^3$ contains air at 16 bar and 42°c. A valve is opened and some air is allowed to blow out to the atmosphere. The pressure of the air in the receiver drops rapidly to 12 bar when the valve is closed. Calculate the mass of air which has left the reciever.

Solution



mass balance

$m_i - m_e = m_2 - m_1$

Energy balance

$$m_i\left(h_i + \frac{v_i^2}{2} + gz_i\right) + Q = m_e\left(h_e + \frac{v_e^2}{2}\right) + m_2u_2 - m_1u_1$$

$PV = mRT$ $\qquad m_e = m_2 - m_1 \qquad (m_i = 0)$

$$m_1 = \frac{P_1V_1}{RT_1}, \quad m_2 = \frac{P_2V_2}{RT_2}$$

→ $P_1 = 16$ bar

$V_1 = 5.5 m^3$

$T_1 = 42°c = 315 K$

$$m_1(kg) = \frac{10^5 N/m^2 \times 16 \times 5.5 m^3}{0.287 J/kgK \times 315K \times 10^3} = 97.34 kg.$$

→ for m_2 (Reversible adiabatically)

$\boxed{PV^\gamma = C}$

$$\left(\frac{P_2}{P_1}\right)^{\gamma - \frac{1}{\gamma}} = \frac{T_2}{T_1} \qquad T_2 = 290K.$$

$P_2 = 12$ bar

$V_2 = 5.5 m^3$

$T_2 = 290 K.$

$$m_2 = \frac{12 \times 100 \times 5.5}{0.287 kJ/kgK \times 290} = 79.3 kg.$$

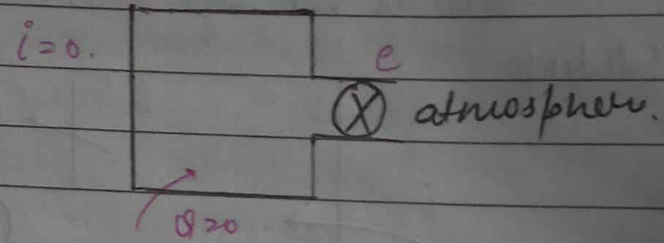
Mass of air left the receiver $= 97.34 - 79.3$ kg $= 18$ kg Ans

Q. 1.6 m^3 dark is filled with air at a pressure of 5 bar and a temp of 100°c. The air is then left out to the atm through a valve. Assuming no heat transfer. determine the work obtainable by utilising ~~air~~ the kinetic energy of the discharge air to run a frictionless turbine.

Solution- mass balance

$\dot{m}_i - m_e = m_2 - m_1$

Energy balance :-



$m_i = 0$ (mass entering)

$W = 0$ (moving part - x)

Energy balance.

$$m_i\left(h_i + \frac{V_i^2}{2} + gz_i\right) + Q = W + m_e\left(h_e + \frac{V_e^2}{2} + gz_e\right) + m_2u_2 - m_1u_1$$

$$m_2\left(h_2 + \frac{V_2^2}{2} + gz_2\right) - m_1\left(h_1 + \frac{V_1^2}{2} + gz_1\right)$$

$$m_1 = \frac{P_1V_1}{RT_1} = \frac{5 \times 10^2 \times 1.6}{0.289 \times (100 + 275)}$$

$R = C_P - C_V = 0.289 \ \frac{kJ}{kg \cdot K}$.

$m_1 = 7.42 \ kg$

$\gamma = \frac{C_P}{C_V} = 1.40$

$$m_2 = \frac{P_2V_2}{PT_2} = 2.35$$

$P_2 =$ 1 bar (atmosphere)

$V_2 = 1.6 m^2$ (container volume - x)

$$\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T_2}{T_1}$$

$$\boxed{T_2 = 235 K}$$

$u_2 = C_v T_2 = 167.085$

$u_1 = C_v T_1 = 265.203$

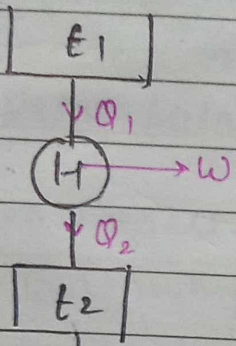
$h_e = C_v T_2 = 167.085$

$m_e = m_1 - m_2 = 5.06$

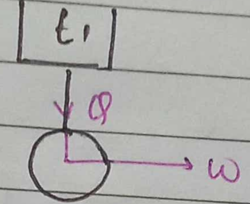
$m_2 u_2 - m_1 u_1 + m_e h_e$

Second Law of Thermodynamics..

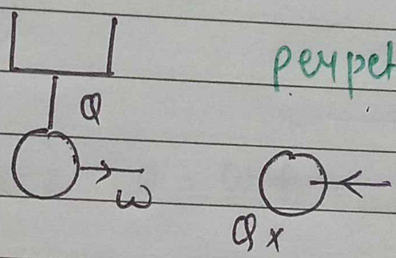
Kelvin Planks Statement – It is impossible ~~to produced~~ for a heat Engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.



$$\boxed{\eta_H = \frac{W}{Q_1}}$$



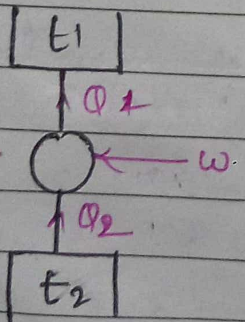
perpetual Motion Machine
PMM2



perpetual Motion Machine
PMM2.

Clausius Statement – It is impossible to construct a device which operating in a cycle, will produce no effect other than the transfer of heat from a cooler body to hotter body.

Heat pump/
Refrigerator.



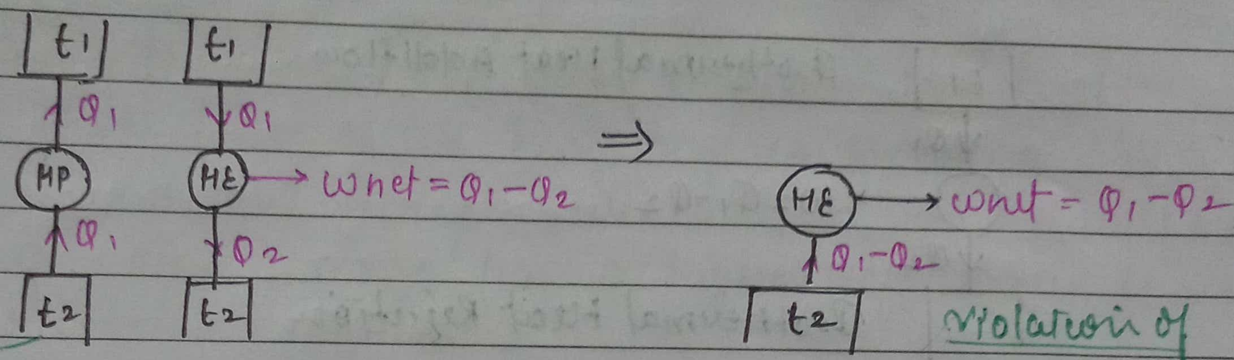
$$\boxed{COP = \frac{Q_2}{W}}$$

Always more than 1

$$\boxed{COP = \frac{Q_1}{W}}$$

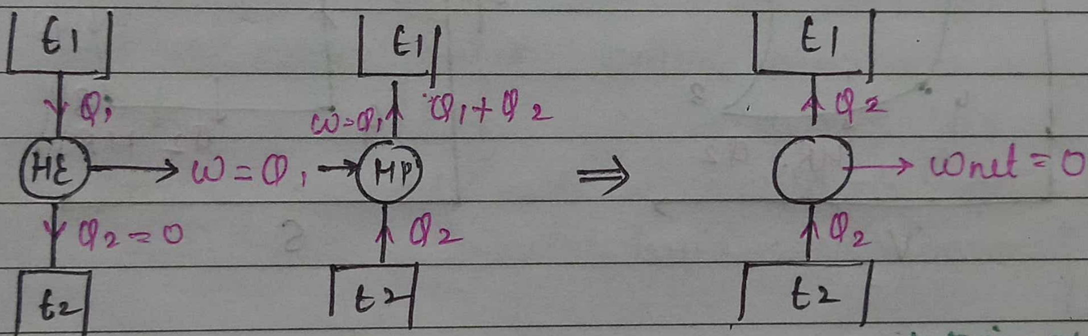
$$\boxed{COP_{Heat\,pump} = 1 + COP_{ref.}}$$

Equivalence of K-P & Clausius Statement -



violation of Clausius statement

violation of K-P.



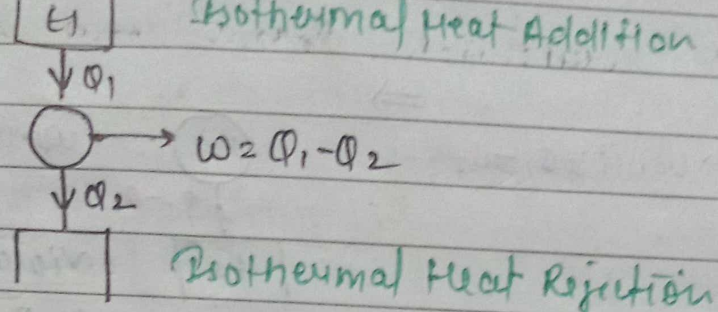
violation of kp

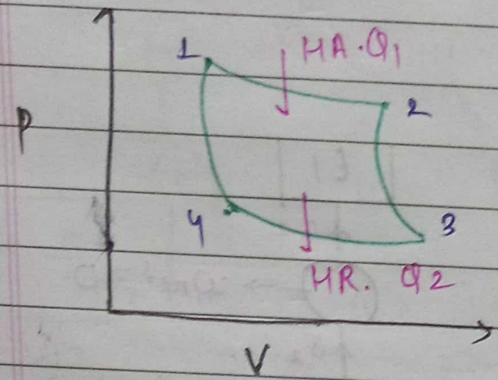
violation of Clausius.

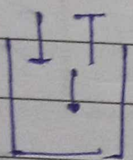
Reversibility & Irreversibility

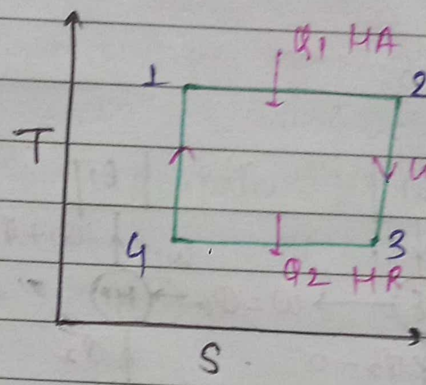
→ Two causes of Irreversibility - ① lack of equillibrium.
② dissipative effect
↳ friction
↳ viscosity

CARNOT CYCLE - four Reversible processes.





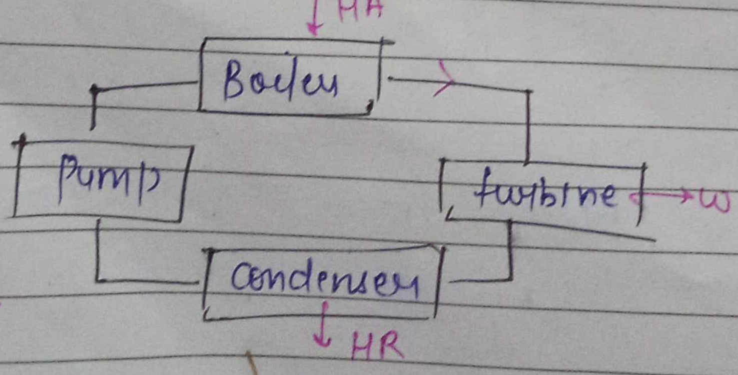




1-2 Heat addition T = C
2-3 Expansion - Rev. Adiabatic process S = const.
3-4 Heat Rejection T = C
4-1 Compression - Rev. Adiabatic process S = C.

$$\eta_{eff} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

→ for 100% efficiency $Q_2 = 0$ not possible.

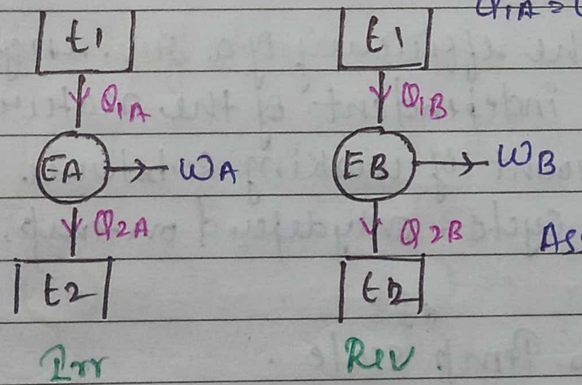


Steady flow.

UNIT-II
Entropy and Availability

① Carnot's theorem / corollary statement and Proof -
It states that of all heat engines operating b/w a given constant temperature and a given constant tempr sink, none has a higher efficiency than a reversible engine.

$Q_{1A} = Q_{1B} = Q_1$



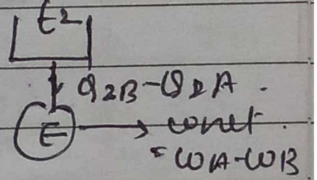
We have to prove - $\eta_B > \eta_A$.

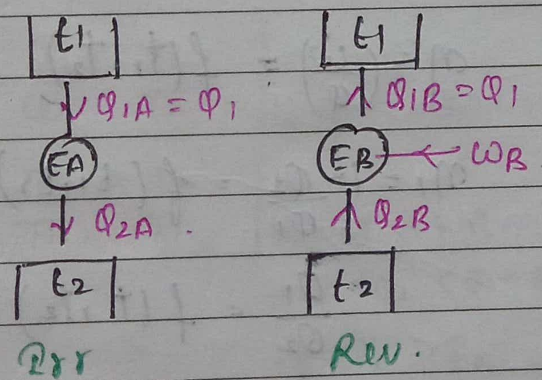
$Q_{1A} = Q_{1B} = Q_1$

Assume $\eta_A > \eta_B$

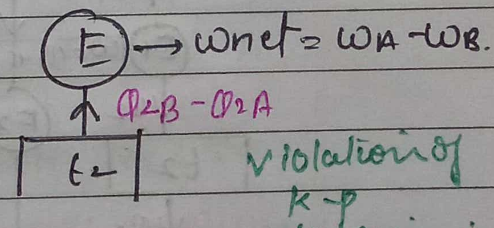
$\Rightarrow \frac{W_A}{Q_{1A}} > \frac{W_B}{Q_{1B}}$

$\Rightarrow W_A > W_B$





$\Rightarrow$



Violation of K-P statement

So Assumption is wrong $\eta_A > \eta_B$

So $\boxed{\eta_B > \eta_A}$ HP.

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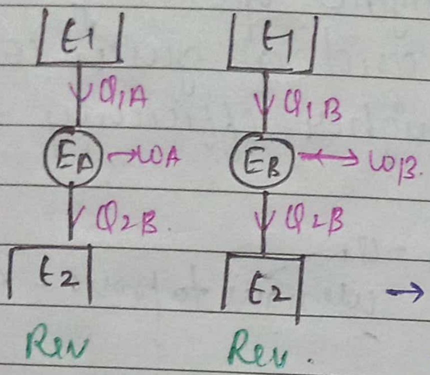
→ Corollary 2

The efficiency of all reversible heat engines operating b/w same tempʳ reservoirs is same.



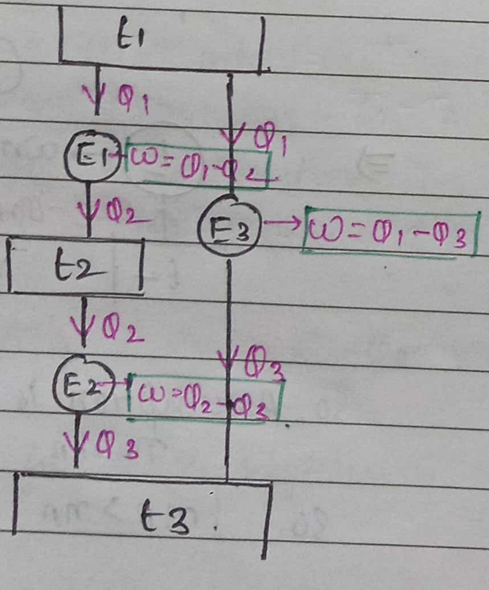
$\eta_B \geq \eta_A$ | $\eta_A \geq \eta_B$

$$\boxed{\eta_A = \eta_B}$$

→ The efficiency of a rev. engine is independent of the nature or amount of working substance undergoing the cycle, only depend on temp.

→ Corollary 3 - Related to Temp scale.

Thermodynamic temp scale.



$$\eta = \left(\frac{W}{Q}\right) = f(t_1, t_2)$$

$$\eta_1 = 1 - \frac{Q_2}{Q_1} = f(t_1, t_2)$$

$$\frac{Q_1}{Q_2} = f(t_1, t_2)$$

$$\frac{Q_2}{Q_3} = f(t_2, t_3)$$

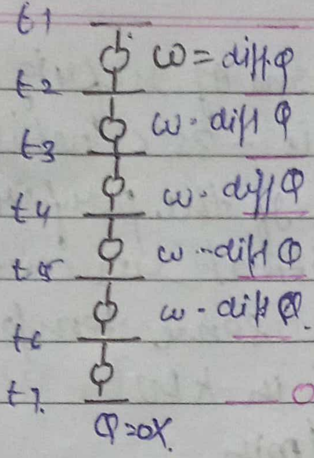
$$\frac{Q_1}{Q_3} = f(t_1, t_3)$$

$$\Rightarrow \frac{Q_1}{Q_2} = \frac{Q_1/Q_3}{Q_2/Q_3} = f(t_1, t_2) = \frac{f(t_1, t_3)}{f(t_2, t_3)} = \frac{\phi t_1 \; \cancel{\phi t_3}}{\phi t_2 \; \cancel{\phi t_3}}$$

$$\Rightarrow \frac{Q_1}{Q_2} = \frac{\phi t_1}{\phi t_2} \Rightarrow \boxed{\frac{Q_1}{Q_2} = \frac{t_1}{t_2}}$$

1st law = E' internal energy.
2 law : Entropy

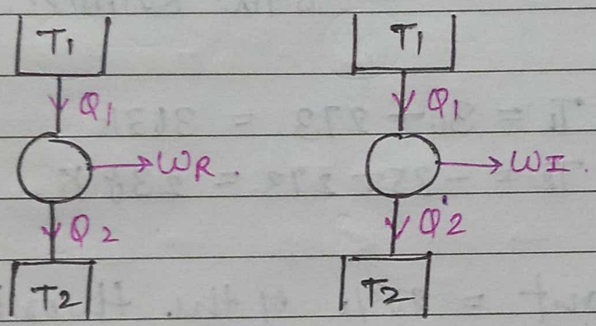
Temperature scale.



diff same.

Actually K = 0K is not possible.

Inequality of clausius :



→ from K-P statement

$\oint Q_{HE} = Q_1 - Q_2 > 0$ OR

$Q_1 - Q_2' > 0$ Rev

$\oint \frac{dQ}{T}\Big|_{HE \cdot R} = \boxed{\frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0}$

$\oint \frac{dQ}{T}\Big|_{HE\ Irr} = \frac{Q_1}{T_1} - \frac{Q_2'}{T_2}$ Irr

$= \boxed{\frac{Q_1}{T_1} - \frac{Q_2'}{T_2} < 0}$

for Rev engine only ⇒ $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

⇒ $\frac{Q_1}{T_1} = \frac{Q_2}{T_1}$

⇒ $\eta_I < \eta_R$

⇒ $\frac{W_I}{Q_1} < \frac{W_R}{Q_1}$

⇒ $W_I < W_R$

⇒ $Q_1 - Q_2' < Q_1 - Q_2$

⇒ $Q_2' > Q_2$

$\boxed{\oint \frac{dQ}{T} \le 0}$ Always

Both Rev & Irr.

↳ Clausius theorem.

Numerical

① $\oint \frac{dQ}{T} = 0$ Rev cycle

② $\oint \frac{dQ}{T} < 0$ Irr / possible cycle.

③ $\oint \frac{dQ}{T} > 0$ Not possible cycle.

Q A fish freezing plant requires 40 tons of refrig- eration. The freezing temperature is $-35°C$ while the ambient temperature is $30°C$. If the performance of the plant is 20% of the theoritical reversed carnot cycle working within the same temp limits Calculate power required in kW
1 ton of refrigeration = 210 kJ/min
Given:-

Solution cooling required = 40 tons = 40×210.
$= 8400$ kJ/min.

⇒ Ambient temp $= T_1 = 30 + 273 = 303K$
⇒ freezing temp $= T_2 = -35 + 273 = 238K$.

⇒ performance of plant = 20% of the theoritical reserved carnot.

⇒ $(COP)_{ref} = \frac{T_2}{T_1 - T_2} = \frac{238}{303 - 238} = 3.66$

⇒ Actual COP $= 0.20 \times 3.66 = 0.732$

$(COP)_{actual} = \frac{\text{cooling req}}{\text{work needed}}$

$0.732 = \frac{8400}{W}$

$W = \frac{8400}{0.732}$ kJ/min

$W = 191.25$ kJ/s / 191.25 kW.

$\boxed{\text{power} = 191.25 \text{ kW}}$

Entropy

$\Rightarrow \int_i^f \frac{dQ_R}{T} = S_f - S_i$ Reversible

$\Rightarrow dQ = T(S_f - S_i)$

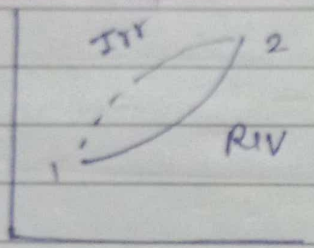
$\boxed{dQ = T\,ds}$

Reversible process.

$S \rightarrow kJ/K$ Entropy

$s = (S/m)$ specific entropy

For Irreversible process –

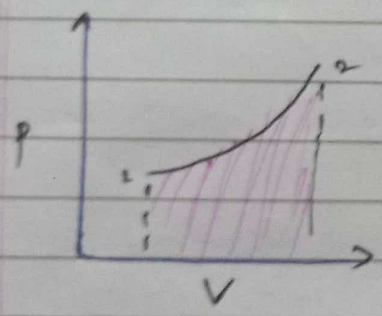


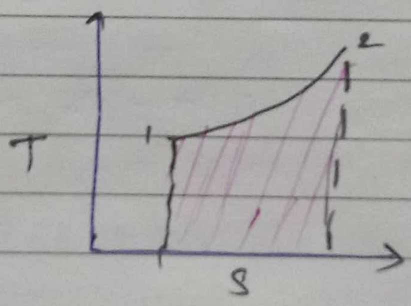
$$\int_1^2 \left(\frac{dQ}{T}\right)_R = (dS)_{Irr} = (ds)_{Rev}$$

$\rightarrow dQ = T\,ds \rightarrow Rev$

$\rightarrow \frac{dQ}{T} < ds \rightarrow Irr.$

$\Rightarrow \boxed{ds \geq \int \frac{dQ}{T}}$





Area under curve $\rightarrow$ $W = p\,dv.$

$dQ = T\,ds$

Entropy Principles –

$\Rightarrow ds \geq \frac{dQ}{T}$

$\Rightarrow$ for adiabatic $\boxed{ds \geq 0}$

$\boxed{ds_{surr} \geq 0}$

$\boxed{ds_{univers.} \geq 0}$ (Isolated)

$ds = 0$ if all rev

$ds > 0$ irreversible

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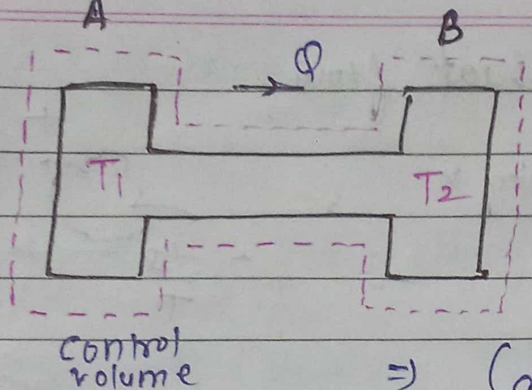
(1) 

Transfer of heat through finite temp diff ds ??

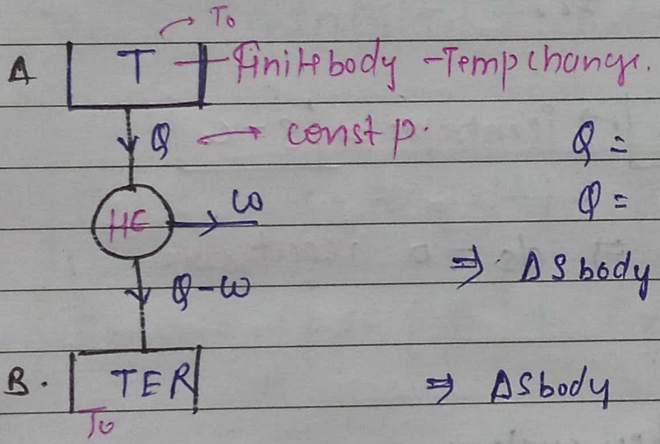
$$dQ = T\,ds \Rightarrow ds = \frac{dQ}{T}$$

$$\Rightarrow (ds)_A = (dQ/T)_A = -Q/T_1$$

$$\Rightarrow (ds)_B = +Q/T_1$$

$$\Rightarrow \Delta S = Q\left(-\frac{1}{T_1} + \frac{1}{T_2}\right) > 0$$

(2) Maximum work obtainable from a finite body & a TER.



$Q = m\,c_p\,dT \rightarrow$ const Pressure.

$Q = c_p(T - T_0)$

$$\Rightarrow \Delta S_{body} = \int_T^{T_0} \frac{dQ}{T} = \int_T^{T_0} \frac{c_p\,dT}{T}$$

$$\Rightarrow \Delta S_{body} = c_p \ln\left(\frac{T_f}{T_i}\right) = c_p \ln(T_0/T)$$

$$\Rightarrow \Delta S_{HE} = 0$$

$$\Rightarrow \Delta S_{TER} = \int \frac{dQ}{T} = \frac{(Q-W)}{T_0}$$

$$\Rightarrow \Delta S \geq 0$$

$$\Rightarrow c_p \ln(T_0/T) + \left(\frac{Q-W}{T_0}\right) \geq 0$$

$$\Rightarrow c_p \ln(T_0/T) \geq \frac{W-Q}{T_0}$$

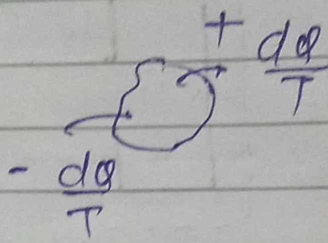
$$\Rightarrow c_p \ln(T_0/T) + \frac{Q}{T_0} \geq \frac{W}{T_0}$$

$$\Rightarrow W \leq c_p T_0 \ln(T_0/T) + c_p(T - T_0)$$

$$W_{max} = c_p T_0 \ln(T_0/T) + c_p(T - T_0)$$

Entropy transfer with heat flow

work flow = entropy ✗.



Entropy generation in a closed system.

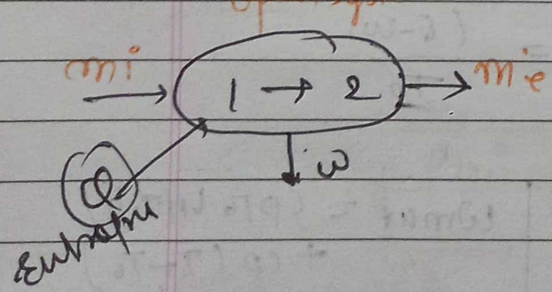
$\Rightarrow \quad ds \geq \frac{dQ}{T}$

$$\boxed{ds = \underbrace{\frac{dQ}{T}}_{\text{Heat transfer}} + \underbrace{s_{gen}}_{\text{Internal Irr}}} \rightarrow \text{positive.}$$

→ Rev adiabatic $\boxed{ds = 0}$ → isentropic ⇒ $\boxed{s = c}$

Irr → $S_{generation} = \left(-\frac{Q}{T}\right) \Rightarrow ds = 0$ isentropic.

Entropy change in open system.



$= \quad m_i - m_e = \frac{dm}{dt}$

$= m_i\left(h_1 + \frac{v_1^2}{2} + gz_1\right) + Q = W + m_e\left(h_2 + \frac{v_2^2}{2} + z_2\right) + \frac{dE}{dz}$

$= \Sigma m_i s_i + \frac{\dot{Q}}{T} + s_{gen} = \Sigma m_e s_e + \frac{\dot{Q}}{T} + s_{gen}$

$$\boxed{\frac{ds}{dz} = \Sigma m_i s_i - \Sigma m_e s_e + \frac{Q}{T} + s_{gen}.}$$

steady state $\quad m_i = m_e = m$

heat transfer - entropy increase due to molecule chaos.

$$\boxed{s_i - s_e + \frac{1}{m}\left(\frac{Q}{T}\right) + \frac{s_{gen}}{m} = 0}$$

Entropy change for a closed system (Rev processes - gases)

1st law $dQ = du + dw$

2nd law → $dQ = Tds$

$$Tds = \underbrace{du}_{mcdT} + \underbrace{dw}_{pdv}$$

$$Tds = c_v dT + p dv$$

$$\int_1^2 ds = \int_{T_1}^{T_2} c_v \frac{dT}{T} + \int_1^2 \frac{P}{T} dv$$

$PV = mRT$

$P/T = R/V$

$$\int ds = \int c_v \frac{dT}{T} + \int \frac{R}{V} dv$$

→ $\boxed{s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln\left(\frac{V_2}{V_1}\right)}$

$\therefore \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$\therefore c_p - c_v = R$

→ $\boxed{s_2 - s_1 = c_p \ln(T_2/T_1) - R \ln(P_2/P_1)}$

→ $\boxed{s_2 - s_1 = c_p \ln\left(\frac{V_2}{V_1}\right) + c_v \ln(P_2/P_1)}$

const volume.

$$dQ = du + \cancel{pdv}^{\,0}$$

$$Tds = c_v dT$$

$$ds = c_v \frac{dT}{T}$$

$$\boxed{s_2 - s_1 = c_v \ln\left(\frac{T_2}{T_1}\right)}$$

const temp

⇒ $dQ = du + pdv$

⇒ $dQ = c_v d\overset{0}{T} + pdv$

⇒ $dQ = pdv$

⇒ $Tds = pdv$

⇒ $ds = P/T\, dv$

⇒ $\int ds = \int \frac{R}{V} dv$

⇒ $\boxed{s_2 - s_1 = R \ln\left(\frac{V_2}{V_1}\right)}$

const pressure

$$dQ = du + pdv$$

$$dQ = c_p dT$$

$$\int Tds = c_p \int dT$$

$$\boxed{s_2 - s_1 = c_p \ln(T_2/T_1)}$$

#. Exergy / Available Energy / Availability / maximum useful work.

[low grade energy]
↓ continuous cycle.

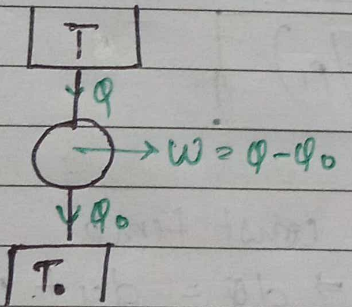
max amount obtained as work, W — **Exergy**.

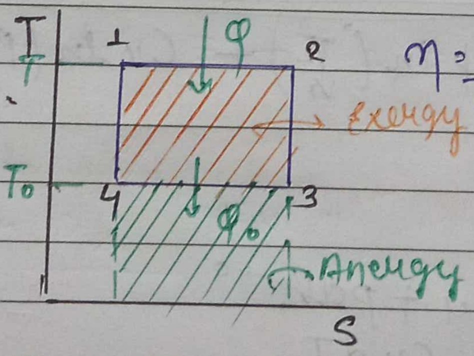
min amount left as low grade energy Q. — **Anergy / unavailable energy / unavailability**

$Q \longrightarrow W$
LGE ✗ HGE
$W \longrightarrow Q$

→ Dead state = system ⟹ surrounding state [work potential = 0 because no gradient]

Exergy / AE →
Energy / AE of heat input in a cycle ⊥



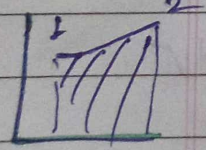


$\eta = \frac{W}{Q}$ $\quad \eta_{max} = \frac{W_{max}}{Q} = \left(1 - \frac{T_0}{T}\right)$

$W_{max} = Q\left[1 - \frac{T_0}{T}\right]$

Exergy of Q at Temp w.r.t to T_0 (surrounding temp)

$$\boxed{W_{max} = Q\left(1 - \frac{T_0}{T}\right)}$$



$\int T\,ds = dQ = Q_1 - Q_2$

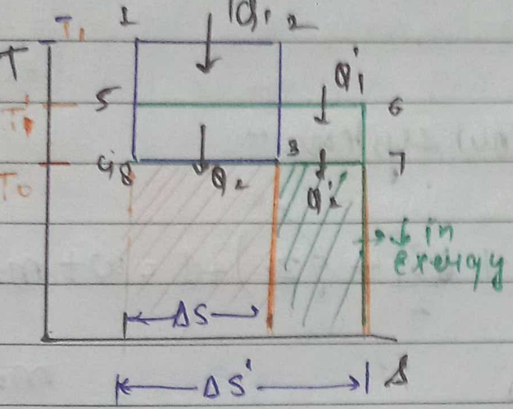
$\oint dQ = \oint dW$

$Q - Q_0 = W$.

$W_{max} = Q - Q_0$

$$\boxed{\text{Anergy} = T_0(S_3 - S_4) = T_0 \Delta S}$$

Decrease in Exergy when heat is transfer through a finite temperature different -



AE of Q_1 at T_1 const T_0

$$Q_1\left(1-\frac{T_0}{T_1}\right) = w_{max}$$

AE of Q_1 at T_1' wrt T_0 =

$$Q_1\left(1-\frac{T_0}{T_1'}\right) = w_{max}.$$

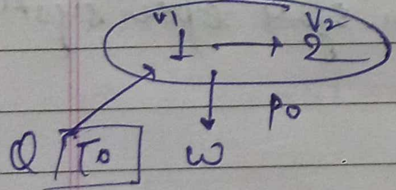
So decrease in AE $= Q_1\left[\frac{T_0}{T_1'} - \frac{T_0}{T_1}\right]$

Note Heat transfer Low temp → work potential ↓ → law of Degradation of quality -

decrease in Exergy $= T_0(\Delta S') - T_0(\Delta S)$
$= Q_2' - Q_2$

Exergy change of a closed system -

Assume rev system.



$$w_{max} = w - p_0(V_2 - V_1) \quad \text{—(A)}$$

↓ w_{useful}

Apply 1st law.

$$Q = \Delta U + W = U_2 - U_1 + W \quad \text{—(1)}$$

$$\Delta S_{sys} + \Delta S_{surr} = 0$$

$$\boxed{S_2 - S_1 + \left(-\frac{Q}{T_0}\right) = 0} \quad \text{—(2)}$$

$$S_2 - S_1 = \frac{Q}{T} + S_{gen}^{\,0} \quad \text{rev.}$$

$$Q = T_0(S_2 - S_1) \quad \text{—(3)}$$

By (1) & (3)

$$T_0(S_2 - S_1) = U_2 - U_1 + W.$$

$$W = U_1 - U_2 + T_0(S_2 - S_1) \text{ - put in (A)}$$

$$W_{max} = U_1 - U_2 + T_0(S_2 - S_1) - P_0(V_2 - V_1)$$

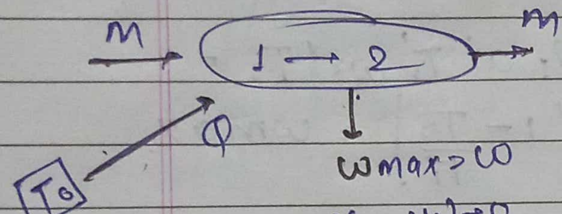
$$W_{max} = (U_1 + p_0V_1 - T_0S_1) - (U_2 + p_0V_2 - T_0S_2)$$

composite property ↳ $\boxed{W_{max} = \Phi_1 - \Phi_2}$ Exergy of closed system

→ Dead state (0) 1 → dead state.

$$\boxed{W_{max} = \phi_1 - \phi_0}$$

Exergy of a steady flow system -



$$m\left(h_1 + \cancel{\frac{V_1^2}{2}} + \cancel{gz_1}\right) + Q = W + m_2\left(h_2 + \cancel{\frac{V_2^2}{2} + gz_2}\right)$$

$W_{max} > W$

$-p(v_2 - v_1) = 0$

rigid.

$m_2 h_2 = H_2$

$m_1 h_1 = H_1$

1st law

$$H_1 + Q = H_2 + W$$

$$H_2 - H_1 = Q - W \quad —(1)$$

2nd law

$$S_2 - S_1 - Q/T_0 = 0$$

$$Q = T_0(S_2 - S_1) \rightarrow (2)$$

$$H_2 - H_1 = T_0(S_2 - S_1) - W$$

$$\boxed{W = (H_1 - T_0 S_1) - (H_2 - T_0 S_2)}$$

considered all para mete.

$$W = \left(H_1 + \frac{mV_1^2}{2} + mgz_1 - T_0 S_1\right) - \left(H_2 + \frac{mV_2^2}{2} + mgz_2 - T_0 S_2\right)$$

$H - T_0 S$

↓

B → keenan function

$$\boxed{W = \psi_1 - \psi_2}$$ Exergy fun. for open system

$B_1 - B_2 / m$

$b_1 - b_2$

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Q A piston cylinder arrangement certains 0.05m^3 of Nitrogen at 1 bar and 280 k. The piston moves inwards and the gas is compressed isothermally and reversibly if untill the pressure becomes 5 bar. Determine the change in entropy. N_2 → perfect gas.

Sol^n –

$P_1 = 1 \text{ bar}$

$P_2 = 5 \text{ bar}$

$T = 280 k$ (const) isothermal

$V_1 = 0.05 m^3$

$\Rightarrow dQ = du + pdv$

$\Rightarrow dQ = \cancel{m c_v dT} + pdv$

$\Rightarrow dQ = pdv$

$\Rightarrow Tds = pdv$

$\Rightarrow \cancel{T}ds = \frac{R\cancel{T}}{V} dv$

$PV = RT$

$P = \frac{RT}{V}$

$\Rightarrow \int ds = \int \frac{R}{V} dv$

$\Rightarrow s_2 - s_1 = R \ln\left(\frac{V_2}{V_1}\right) \times m$

$\Rightarrow P_1 V_1 = RT_1$ $\quad T_1 = T_2$

$P_2 V_2 = RT_2$

$P_1 V_1 = P_2 V_2$

$\Rightarrow s_2 - s_1 = R \ln\left(\frac{P_1}{P_2}\right) \times m$

$\Rightarrow s_2 - s_1 = 0.297 \ln\left(\frac{1}{5}\right) \times m$

$R = \frac{R_u}{M} = \frac{8.314}{28}$ KJ/Kmol Kelvin

$\Rightarrow \boxed{s_2 - s_1 = -0.028}$

$R = 0.297$ KJ/KgJ

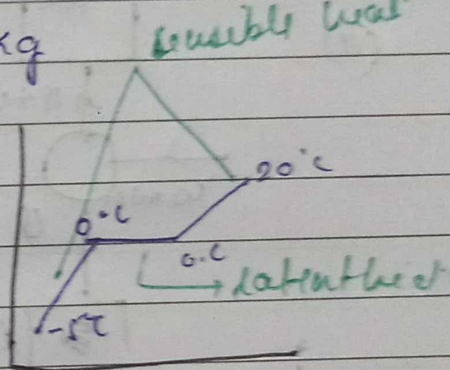
$m = \frac{P_1 V_1}{RT_1} = \frac{10^2 \times 0.05}{0.297 \times 280}$

$m = 0.060125$

Q 1kg of ice at $-5^\circ C$ is exposed to atmosphere which is at $20^\circ C$ the ice melt and comes to thermal equillibrium with the atmosphere.
i) determine the entropy increase of the universe
ii) what is the min amount of work necessary to convert the water necessity to convert $-5^\circ C$.

→ $Cp_{ice} = 2.093$ KJ/kgK.

→ latent heat of fusion of ice = 333.3 KJ/kg



→ m = 1kg
$-5^\circ C \longrightarrow 20^\circ C$

$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$

ice

Q

surr
20°C
293k

$\Delta S_{surr} = -Q/T_0$

$Q = cp_{ice}(0-(-5)) + 333.3 \text{ KJ/kg} + cp_{water}(20^\circ - (0))$

$Q = 2.093 \times (5) + 333.3 + 4.187 \text{ KJ/kgk} (20)$

$Q = 427.5$ KJ

$\Delta S_{surr} = -Q/T_0 = -1.46$ KJ/k.

$\Rightarrow \dfrac{dQ}{T} = ds$

$\Rightarrow \dfrac{mcpdT}{T} = ds$

$\Rightarrow ds = mcp\ln(T_2/T_1)$

$\Delta S_1 = Cp\ln\left(\dfrac{273}{268}\right)$

$\Delta S_2 = \dfrac{Q}{T} = \dfrac{\text{latent heat}}{273}$ → 333.3

$\Delta S_3 = Cp\ln\left(\dfrac{293K}{273K}\right)$

Q 1kg of ice at $-5°c$ is exposed to atmosphere which is at $20°c$ the ice melt and comes to thermal equillibrium with the atmosphere.

i) determine the entropy increase of the universe

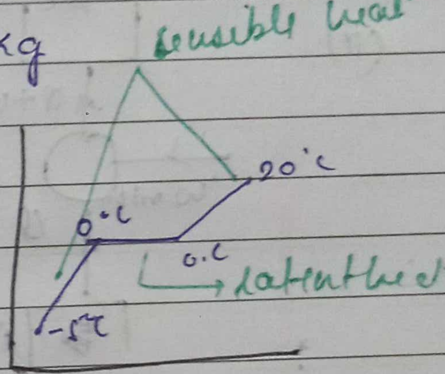
ii) what is the min amount of work necessary to convert the water necessity to convert $-5°C$.

→ $Cp_{ice} = 2.093\ KJ/kgK$.

→ latent heat of fusion of ice = $333.3\ KJ/kg$

→ $m = 1kg$

$-5°C \longrightarrow 20°C$



$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$

ice

Q

surr
20°C
293k

$\Delta S_{surr} = -Q/T_0$

$Q = cp_{ice}(0-(-5)) + 333.3\ KJ/kg + Cp_{water}(20°-(0))$

$Q = 2.093 \times (5) + 333.3 + 4.187\ KJ/kgk\ (20)$

$Q = 427.5\ KJ$

$\Delta S_{surr} = -Q/T_0 = -1.46\ KJ/k$.

$\Rightarrow \dfrac{dQ}{T} = ds$

$\Rightarrow \dfrac{mcpdT}{T} = ds$

$\Rightarrow ds = mcp\ ln(T_2/T_1)$

$\Delta S_1 = Cp\ ln\left(\dfrac{273}{268}\right)$

$\Delta S_2 = \dfrac{Q}{T} = \dfrac{\text{latent heat}}{273}$ → 33.3

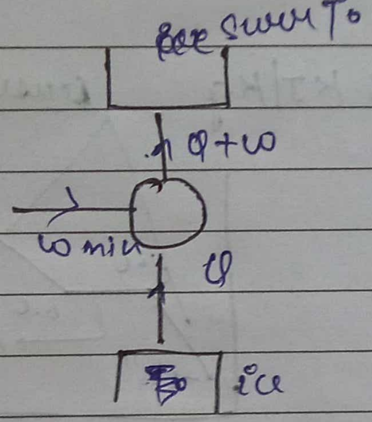
$\Delta S_3 = Cp\ ln\left(\dfrac{293\ K}{273K}\right)$

$$\Delta S_{sys} = \Delta S_1 + \Delta S_2 + \Delta S_3$$

$$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$$

$$\Delta S_{universe} = 0.0949 \text{ kJ/k}$$

(ii)



$$\Delta S_{univer} = \Delta S_{sys} + \Delta S_{surr} \geq 0$$

$$\text{min work} = 0$$

$$\Delta S_{sys} + \frac{Q+w}{T_0} = 0$$

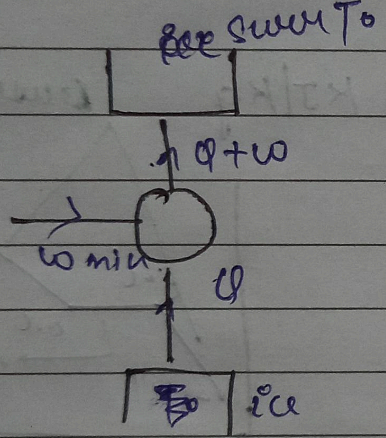
-ve

$$\Delta S_{sys} = \Delta S_1 + \Delta S_2 + \Delta S_3$$

$$\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr}$$

$$\Delta S_{universe} = 0.0949 \text{ kJ/k}$$

(ii)



$$\Delta S_{univers} = \Delta S_{sys} + \Delta S_{surr} \geq 0$$

min work $= 0$

$$\underset{-ve}{\Delta S_{sys}} + \frac{Q+W}{T_0} = 0$$